

Unsupervised Anomaly Detection for Ship Engine Monitoring

D. J. McCann, BSc

28th December 2025

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Problem Statement

The company operates a fleet of ships equipped with onboard sensors that monitor key engine parameters. An anomaly detection system is required to identify abnormal engine behaviour and enable timely safety intervention or preventative maintenance before extreme conditions or mechanical failure occur. Such a system would reduce safety risk, minimise unplanned downtime, and mitigate financial losses arising from operational disruption.

Methods

Domain Context

Domain guidance indicates that anomalous engine states typically arise from combinations of extreme sensor values rather than isolated deviations. Consequently, an anomaly prevalence of approximately 1–5% of observations is considered operationally realistic for this system.

The Dataset

The dataset contains 19,535 observations across six engine features, with no missing, duplicate, or invalid values detected. Two observations exhibit extreme coolant temperatures ($>6\sigma$) inconsistent with expected thermodynamic coupling; these are retained pending domain validation and are displayed in Table 1.

Row Number	Engine RPM	Oil Pressure	Fuel Pressure	Coolant Pressure	Oil Temperature	Coolant Temperature
3242	760	4.9	2.6	1.7	77.9	118.4
10663	455	2.0	7.8	1.6	76.5	195.5

Table 1 - Suspect Errors

Distribution Analysis

Several variables exhibit non-Gaussian distributions; however, when restricted to the central operating regime, coolant temperature and fuel pressure approximate normality, indicating that non-Gaussian behaviour is primarily driven by extreme tail values.

Correlation Structure

While correlations are weak overall, moderate inverse correlations (approximately -0.2 to -0.3) emerge when analysis is restricted to extreme observations ($>3\sigma$), indicating increased coupling under stress.

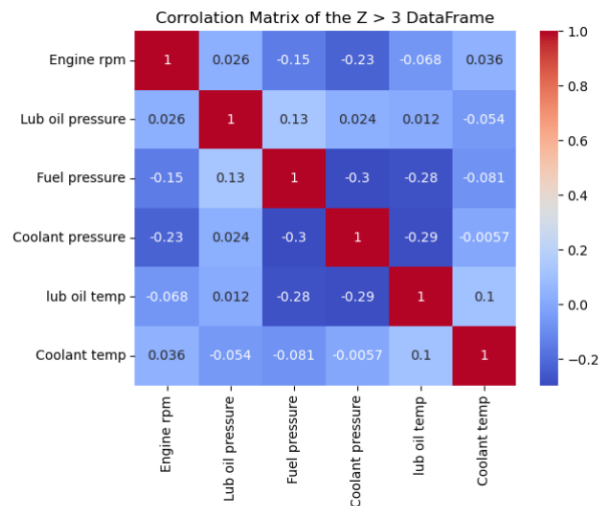


Figure 1 - Correlation Heat Map of Engine Variables for Extreme Conditions ($Z > 3$)

Interquartile Range (IQR)

IQR boundaries were defined relative to the upper and lower quartiles ($Q1$, $Q3$):

- Lower: $Q1 - (1.5)IQR$
- Upper: $Q3 + (1.5)IQR$

Each observation was assigned an outlier count equal to the number of exceeded features for that row, allowing anomaly prevalence to be evaluated as a function of multi-feature deviation.

Outlier Boundary	Outlier Count	Percentage of Data
1	4636	23.73%
2	422	2.16%
3	11	0.06%

Table 2 - IQR Outlier Detection for Given Boundary

Model Evaluation (Unsupervised)

Without labelled ground truth, performance was evaluated using a centroid-based geometric separation metric between predicted normal and anomalous observations.

One-Class Support Vector Machine (OCSVM)

A One-Class Support Vector Machine with an RBF kernel was trained on standardised features. Hyperparameters were tuned to achieve an anomaly rate within the expected 1–5% range, and model performance was assessed using a centroid-based separation metric.

Isolation Forest (IF)

An Isolation Forest was trained on the unscaled feature space, with contamination and tree parameters varied to obtain anomaly rates within the expected range. Model performance was evaluated using the same separation metric for consistency.

Model Visualisation & Interpretation

Principal Component Analysis (PCA)

Model behaviour was visualised using PCA, with points coloured by binary anomaly labels and the continuous decision function score.

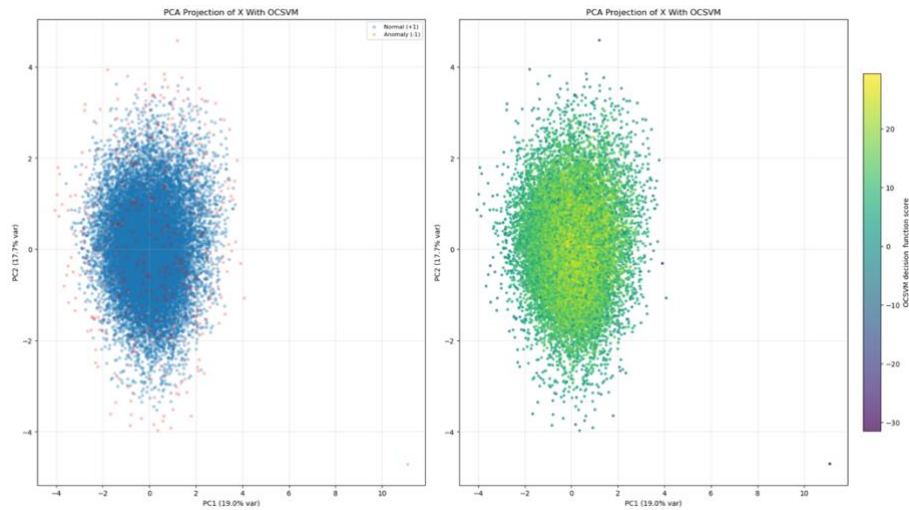


Figure 2 – PCA Projection of the OCSVM Results

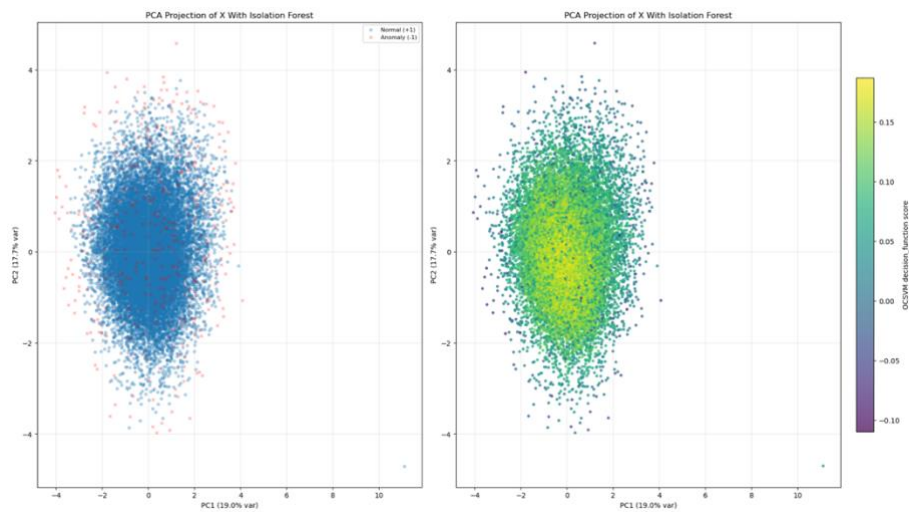


Figure 3 – PCA Projection of the Isolation Forest Results

The PCA projections show anomalies dispersed within dense regions, with noisy decision-score gradients, though smoother for Isolation Forest, and poor confinement of extreme values to geometric boundaries. This indicates limited alignment with linear variance structure and suggests the presence of underlying nonlinear relationships.

Uniform Manifold Approximation and Projection (UMAP)

UMAP was implemented to visualise model predictions and decision scores, to preserve local, potentially nonlinear, neighbourhood structure.

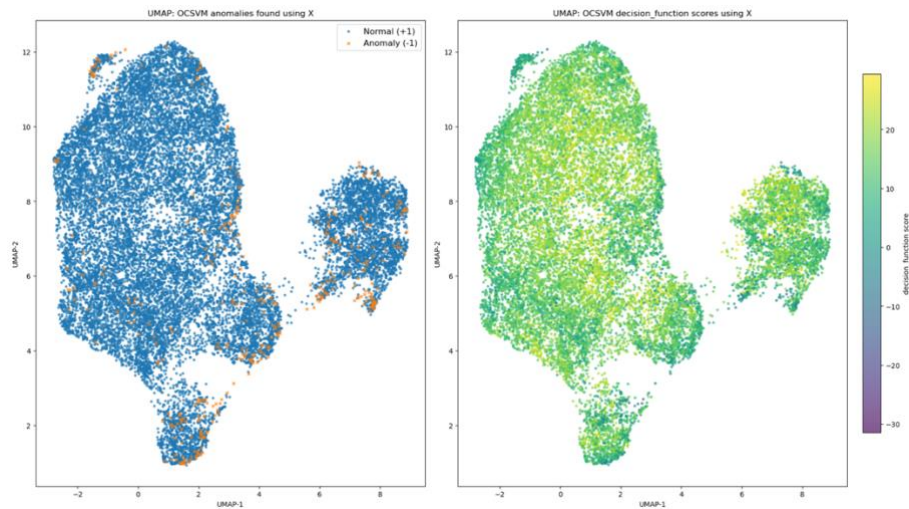


Figure 4 – UMAP Projection of the OCSVM Results

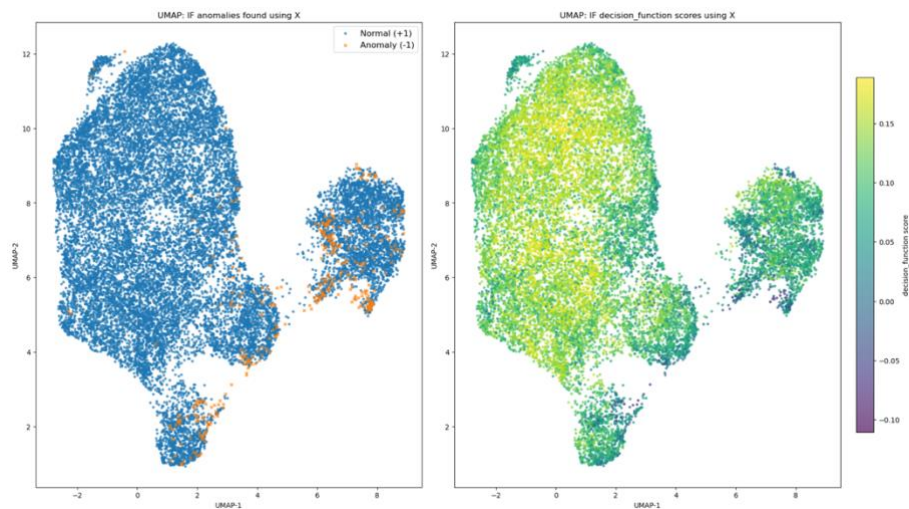


Figure 5 - UMAP Projection of the Isolation Forest Results

UMAP reveals clearer structure and smoother decision-function gradients than PCA, with anomalies more frequently aligned to low-density regions and manifold boundaries, particularly for Isolation Forest. This supports the presence of nonlinear structure and explains the limited interpretability of linear PCA projections.

Results

IQR

Requiring at least two features per observation to exceed the IQR threshold identified 422 outliers (2.16% of the dataset), supporting the view that anomalous engine behaviour is better characterised by simultaneous multi-sensor deviations than isolated univariate extremes.

OCSVM

The highest-performing OCSVM configuration ($\gamma = 0.2$, $\eta = 0.02$) identified 389 anomalies (1.99% of the dataset). While PCA shows poor alignment between the decision boundary and linear variance structure, UMAP reveals smoother gradients and clearer boundary structure, indicating that normal engine behaviour is governed by nonlinear interactions.

IF

The highest-performing Isolation Forest model identified 391 anomalies (2.00% of the dataset). Under nonlinear embedding, the decision-score distribution appears smoother and more spatially coherent than for OCSVM, suggesting that the isolation-based mechanism captures local irregularities in the feature space more robustly when complex, nonlinear interactions are present.

Summary

Across all methods, models were tuned to achieve anomaly rates within the 1–5% range (Table 3) and selected based on the separation metric. While individual models differed in their specific predictions, a subset of observations was flagged across methods.

Predicted by IQR	Predicted by OCSVM	Predicted by Isolation Forest
422 (2.16%)	389 (1.99%)	391 (2.00%)

Table 3 – Predictions made by each model

As shown in Table 4, 438 observations were identified by only one model, suggesting low-confidence anomalies. In contrast, 214 observations were flagged by two models and 113 by all three, representing the highest-confidence anomalies. This pattern indicates that a small core of observations exhibits feature behaviour sufficiently atypical to be detected under multiple, fundamentally different anomaly detection paradigms.

Predicted by One Model	Predicted by Two Models	Predicted by Three Models
438	214	113

Table 4 - Confluence of Anomalies Detected Between Models

This model agreement provides a practical basis for prioritisation, with high-confluence anomalies warranting immediate investigation and lower-confluence cases suitable for monitoring.

Next Steps

Domain Expert Validation

Confirm whether the extreme coolant-temperature observations represent genuine operating conditions or sensor artefacts and assess the operational relevance of detected anomalies.

Further feature Engineering

Apply symbolic regression to identify interpretable nonlinear relationships between engine parameters and construct physics-informed features.

Ensemble Approach

Combine model outputs and prioritise anomalies based on cross-model agreement to improve robustness and reduce false positives.